Practice Set 20 SAQ Sheet – Short Answer Practice

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Short-answer practice sheet compiled from handwritten study lists.

Q3a. Growing teratoma syndrome

Type: SAQ (10 marks)

Definition

- GTS (growing teratoma syndrome; Logothetis, 1982): metastatic masses of an NSGCT (non-seminomatous germ cell tumour) **enlarge during or after chemotherapy while AFP (alpha-fetoprotein) and hCG (human chorionic gonadotrophin) are normal or falling**. Resection shows **mature teratoma only**.
- Uncommon: about 2–8% of metastatic NSGCT.

Pathogenesis

- Teratoma is **histologically benign but genetically malignant** (i(12p), aneuploidy) and **chemoresistant**.
- Chemotherapy kills the sensitive elements (embryonal, yolk sac), so markers fall; the teratoma keeps growing, mostly by enlarging cysts. Chemotherapy-induced differentiation is a second theory.
- Main risk factor: teratoma in the orchidectomy specimen.

Clinical features

- Retroperitoneum commonest; also lung, mediastinum, supraclavicular nodes.
- Often silent; or compression: ureteric obstruction, IVC (inferior vena cava) compression or thrombosis, bowel obstruction, back pain.

Evaluation

- Markers before every cycle; **rising markers mean viable cancer, not GTS**.
- CT (computed tomography): enlarging, well-defined, **cystic** mass, sometimes with fat or calcification.
- **FDG-PET (fluorodeoxyglucose positron emission tomography) has no role**: teratoma is not FDG-avid.
- **Differential:** progressing viable GCT (germ cell tumour; markers rise); malignant transformation (somatic sarcoma or carcinoma); necrosis or fibrosis (mass shrinks).

Management

- **Stop chemotherapy**; do not escalate to salvage chemotherapy (ineffective).
- **Prompt complete surgical resection**: PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection), with resection of every other site; adjacent vessel, kidney or bowel resection may be needed. Delay lets the mass become unresectable.
- Radiotherapy is ineffective.
- **Outcome:** favourable after complete resection (long-term survival 75–90% after resection of residual teratoma). Incomplete resection leads to regrowth, **malignant transformation** (rhabdomyosarcoma commonest) or **late relapse**.
- **Also know:** CDK4/6 (cyclin-dependent kinase 4/6) inhibitors studied for unresectable teratoma; lifelong CT and marker follow-up.

Diagram to draw: graph of mass size rising while marker curve falls during chemotherapy cycles.

> **Key points to remember:**
> - Enlarging mass plus normal or falling markers equals GTS.
> - Teratoma is chemoresistant and not FDG-avid.
> - Treatment is complete surgical resection, not more chemotherapy.

Q3b. Stone complexity scores

Type: SAQ (10 marks)

Purpose

- Grade stone and kidney complexity, chiefly to **predict stone-free rate (SFR) after PCNL (percutaneous nephrolithotomy)**. Used for counselling, choosing the surgeon, auditing and comparing series.

Main scores

Score	Variables and scoring	Predicts
Guy's Stone Score (Thomas, 2011)	**Grade I:** single stone in mid or lower pole, or pelvis, simple anatomy. **II:** single upper-pole stone, or multiple stones with simple anatomy, or single stone with abnormal anatomy. **III:** multiple stones with abnormal anatomy, calyceal diverticulum stone, or partial staghorn. **IV:** complete staghorn, or any stone in spina bifida or spinal injury	SFR on KUB (kidney, ureter, bladder radiograph); not complications
S.T.O.N.E. nephrolithometry	**S**ize (area), **T**ract length (≤100 or >100 mm), **O**bstruction (hydronephrosis), **N**umber of involved calyces, **E**ssence (density, ≤950 or >950 HU [Hounsfield units]); total 5–13	SFR, blood loss, operative time, stay; needs CT (computed tomography)
CROES nomogram (Clinical Research Office of the Endourological Society)	Stone burden, location, staghorn, number, prior treatment, **annual centre volume**	SFR; only multicentre derivation, but SFR on KUB
S-ReSC (Seoul National University Renal Stone Complexity)	Number of **9 sites** involved on CT: pelvis, upper and lower major calyces, anterior and posterior minor calyces of the three groups; score 1–9	SFR; not complications
RUSS (Resorlu–Unsal Stone Score)	1 point each: stone >20 mm; lower-pole stone with infundibulopelvic angle <45°; stones in more than one calyx; abnormal anatomy; score 0–4	SFR after **RIRS** (retrograde intrarenal surgery)

Limitations

- Each score added what the previous lacked: anatomy, then CT depth and density, then centre factors, then calyceal distribution.
- Mostly predict clearance, **not complications**. Meta-analysis: heterogeneous; no single generalisable score.
- **Also know:** Triple D score predicts SWL (shock wave lithotripsy) success, not PCNL.

Diagram to draw: kidney outline showing the 9 S-ReSC sites.

> **Key points to remember:**
>
> - Guy's: grades I–IV; complete staghorn or spina bifida is grade IV.
> - S.T.O.N.E.: five variables, score 5–13.
> - CROES includes centre volume; RUSS is for RIRS.

Q3c. Calyceal diverticulum – classification and management

Type: SAQ (10 marks)

Definition

- Cystic cavity in the renal cortex, **lined by transitional epithelium**, connected to a calyx (or pelvis) by a **narrow neck**. Fills passively with urine.
- About 4.5 per 1000; equal in children and adults; no side or sex preference.

Classification

	Type I	Type II
Frequency	Commoner	Less common
Connects to	Minor calyx (usually upper pole)	Renal pelvis or major calyx
Size, symptoms	Small, usually silent	Larger, often symptomatic

Aetiology

- **Congenital** (favoured): persistent third- and fourth-generation ureteric bud branches.
- **Acquired:** cortical abscess draining into a calyx, infundibular stenosis, trauma, reflux.

Clinical features

- Stasis: **stones in up to 40%**, milk of calcium, infection, pain, haematuria.

Evaluation

- Ultrasound mimics a simple cyst. **CT (computed tomography) urography with delayed images shows contrast pooling**; milk of calcium layers and shifts with position.
- Retrograde pyelography maps the neck before intervention.
- **Differential:** hydrocalycosis (dilated major calyx from infundibular obstruction), megacalycosis, simple cyst.

Management

- **Asymptomatic:** observe with ultrasound.
- **Treat** pain, recurrent infection, haematuria, abscess, symptomatic stones.

Approach	Technique	Best for
Ureteroscopy	Find the ostium, laser-incise or balloon-dilate the neck, remove stones	Upper or mid-pole, small stones
PCNL (percutaneous nephrolithotomy)	Direct puncture onto stone, clear stones, fulgurate lining, dilate or create neck	Posterior, lower-pole, larger stones; long-term results good
Laparoscopic or robotic	Unroof, remove stones, close neck, obliterate cavity, pack with fat	Anterior, thin-parenchyma, failed endoscopy
Partial nephrectomy	Excision	Rare; destroyed segment

- Endoscopy keeps the cavity draining by widening the neck; laparoscopy abolishes it by closing the neck.
- **Also know:** SWL (shock wave lithotripsy) gives symptom relief but low stone clearance; fulguration of lining is debated.

Diagram to draw: kidney with upper-pole diverticulum, narrow neck to calyx, stone inside.

> **Key points to remember:**
>
> - Transitional lined, narrow neck; type I calyceal, type II pelvic.
> - Stones in up to 40%; delayed CT shows pooling.
> - Ureteroscopy or PCNL first; laparoscopy for anterior or failed cases.

Q3d. Transplant renal artery stenosis

Type: SAQ (10 marks) (vascular complications of transplantation also answered in 2025-Winter Paper 1, Q3d)

Definition and incidence

- TRAS (transplant renal artery stenosis): narrowing of the graft artery or its inflow; **commonest vascular complication (1–23%)**, usually 3 months to 2 years after transplant.

Types and causes

- **Anastomotic:** suture technique, end-to-end anastomosis to the internal iliac artery, no aortic (Carrel) patch.
- **Post-anastomotic (distal):** clamp or perfusion-cannula intimal injury, kinking, rejection.
- **Pre-anastomotic ("pseudo-TRAS"):** recipient iliac atherosclerosis.
- **Also know:** CMV (cytomegalovirus) infection, delayed graft function, deceased donor.

Pathophysiology and clinical features

- Reduced graft perfusion activates the **renin–angiotensin–aldosterone system**.
- **Refractory hypertension**, rising creatinine, fluid retention, **flash pulmonary oedema**; graft bruit.
- Clue: creatinine rises sharply after an ACE (angiotensin-converting enzyme) inhibitor.

Evaluation

- **Doppler ultrasound first:** high peak velocity at stenosis (about >200–250 cm/s), and distal **parvus-tardus** waveform with low resistive index.
- MRA (magnetic resonance angiography) or CO_2 angiography to spare contrast.
- **Contrast angiography is the gold standard** and allows treatment at once. Iodinated contrast risks nephrotoxicity; gadolinium is avoided at low GFR (glomerular filtration rate).
- **Differential of graft dysfunction with hypertension:** rejection, calcineurin-inhibitor toxicity, ureteric obstruction.

Management

- **Medical:** for mild stenosis with stable function: blood pressure control (calcium-channel blockers), statin, antiplatelet; avoid or monitor ACE inhibitors.
- **Percutaneous angioplasty with or without stent: treatment of choice** for significant stenosis (usually >50–60%). High technical success; reverses hypertension and graft dysfunction. Stents reduce restenosis.
- **Surgical revision** (resection and re-anastomosis, interposition graft, bypass) if angioplasty fails or anatomy unsuitable; risks graft loss.
- Follow-up Doppler after angioplasty detects restenosis; untreated TRAS ends in graft loss.

Prevention

- Anastomose away from atheroma; avoid clamp injury and kinking; use a Carrel patch where possible.

Diagram to draw: Doppler waveforms: normal and parvus-tardus distal to stenosis.

Key points to remember:

- Commonest vascular complication; presents months later.
- Refractory hypertension plus rising creatinine: Doppler first.
- Angioplasty or stent is first-line; angiography is gold standard.

Q3e. Pelvic lipomatosis

Type: SAQ (10 marks)

Definition

- Rare benign **overgrowth of non-encapsulated mature fat** in the perivesical and perirectal spaces, compressing the bladder, ureters, rectum and pelvic veins.
- Mostly men aged 20–60; associated with obesity and hypertension. Cause unknown; obesity alone does not explain it (weight loss rarely shrinks the fat). Chronic inflammation and venous obstruction are proposed.

Clinical features

- LUTS (lower urinary tract symptoms), haematuria, recurrent UTI (urinary tract infection), suprapubic or perineal pain, constipation.
- Leg oedema or venous thrombosis from venous compression.
- **Hydronephrosis and renal failure** from ureteric compression in a substantial minority.
- **Proliferative cystitis (cystitis glandularis)** is common; it can transform to **adenocarcinoma**.

Evaluation

- **IVU (intravenous urography) or cystogram:** elevated, elongated **"pear-" or "teardrop-shaped" bladder**; medially displaced pelvic ureters; high prostate.
- **Barium enema:** elongated, narrowed rectosigmoid.
- **Ultrasound:** echogenic pelvic fat; hydronephrosis.
- Medial deviation of **both** pelvic ureters also occurs in retroperitoneal fibrosis and after abdominoperineal surgery.
- **CT (computed tomography) or MRI (magnetic resonance imaging): diagnostic**; homogeneous fat density surrounding pelvic organs, separates it from tumour.
- **Cystoscopy:** elevated trigone, long prostatic urethra (rigid scope may be difficult); biopsy for cystitis glandularis.
- **Differential of pear-shaped bladder:** pelvic haematoma, lymphadenopathy, liposarcoma, iliopsoas hypertrophy, venous collaterals, pelvic fibrosis.

Management

- **Surveillance** (often progressive): ultrasound for hydronephrosis, renal function, periodic cystoscopy for cystitis glandularis.
- Weight loss, antibiotics and steroids give limited or no benefit; COX-2 (cyclo-oxygenase-2) inhibitors for cystitis glandularis are case reports only.
- **Ureteric obstruction:** stents or nephrostomy; ureteric reimplantation (difficult, dense fat).
- **Advanced or failed cases:** urinary diversion; cystectomy if adenocarcinoma or unmanageable bladder.
- Surgical excision of fat is difficult: no tissue planes, heavy bleeding.
- Resect cystitis glandularis transurethrally and treat obstruction or inflammation.

Diagram to draw: cystogram outline of pear-shaped bladder with medially deviated ureters.

> **Key points to remember:**
>
> - Pear-shaped bladder with medial ureters; CT shows fat.
> - Cystitis glandularis common; adenocarcinoma risk.
> - Watch for hydronephrosis; diversion if obstruction progresses.

Q3f. Cancer in augmented bladder

Type: SAQ (10 marks)

Introduction

- Bladder augmentation (enterocystoplasty) is mainly for neurogenic bladder. **Earliest reported tumour 4 years after surgery**, usually after decades; tumours are often **aggressive and advanced at diagnosis**.
- German registry (Kälble, 2011): secondary malignancy after cystoplasty **1.6%**, versus ureterosigmoidostomy 2.6% and ileal conduit 0.02%.

Pathology

- **Adenocarcinoma** commonest, usually near the **enterovesical anastomosis**; also urothelial carcinoma (aggressive, often metastatic), signet-ring and other types.
- **Nephrogenic adenoma**: benign, probably the commonest tumour after cystoplasty.

Pathogenesis

- **N-nitroso compounds** from bacteria in stagnant, infected urine (probably enhancers).
- Chronic inflammation at the anastomosis releases growth factors that drive proliferation; perianastomotic urothelium is genetically unstable.
- The underlying disease may itself carry risk: bladder cancer in spina bifida is about 2–4%, mostly stage III or IV, and non-gastric augmentation did not appear to increase that risk.
- Ureterosigmoidostomy is the extreme model: urothelium beside colonic mucosa bathed in faeces; mean latency about 26 years.

Risk factors

- **Gastrocystoplasty** (highest among augments), **exstrophy**, immunosuppression after transplant, smoking, chronic infection, long latency.

Clinical features

- **Gross haematuria**, recurrent UTI (urinary tract infection), pelvic pain, new incontinence, mass. Neurogenic patients may notice symptoms late.

Surveillance

- Chapters differ: Campbell bladder-surgery chapter advises **endoscopy yearly from 10 years**; paediatric chapter: no consensus, not shown cost-effective (Mayo group: unnecessary). Higher-risk groups benefit most.
- Investigate any haematuria promptly: cystoscopy with biopsy, cytology, CT (computed tomography) urography.
- **Prevention:** regular clean intermittent catheterisation, daily bladder irrigation (less stasis, mucus and stones), treat urea-splitting infection, stop smoking, lifelong transitional follow-up.

Management

- **Radical cystectomy with excision of the augment** and urinary diversion (conduit or continent pouch), with lymphadenectomy.
- Chemotherapy as for histology; prognosis poor because of late stage.
- **Also know:** stones, perforation, metabolic acidosis also occur after augmentation.

Diagram to draw: clam ileocystoplasty with tumour at the enterovesical suture line.

Key points to remember:

- Adenocarcinoma at the anastomosis; earliest at 4 years.
- Gastric segment, exstrophy, immunosuppression: highest risk.
- Haematuria needs cystoscopy; treatment is cystectomy.

Rapid Revision Sheet – Practice Set 20 SAQ Sheet

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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